PCF8576D Alternate LCD Driver

Current LCD Driver: PCF8576D
IC Package: TSSOP56
Protocol: I2C
LCD Type est.: TN LCD
Usage: Six 7-segment + PSI, KPA, %, BAT, LPM, 2 indicators.

Driver Description:

- Single chip LCD controller and driver
- Selectable backplane drive configuration: static or 2, 3, 4 backplane multiplexing
- 40 × 4-bit RAM for display data storage (25 x 3 used).
- Selectable display bias configuration: static, 1⁄2, or 1⁄3.
- Auto-incremented display data loading across device subaddress boundaries.
- Integrated Oscillator.
- Independent supplies possible for LCD and logic voltages.
- Wide power supply range: from 1.8 V to 5.5 V
- Wide logic LCD supply range:
    - From 2.5 V for low-threshold LCDs
    - Up to 6.5 V for high-threshold twisted nematic LCDs
- 400 kHz I2C-bus interface

Preferred Alternative

Choice# 1 LCD Driver: PCF8576CT/1,518
IC Package: VSOP-56
Protocol: I2C
Link: PCF8576CT/1,518
Quantity: DigiKey: 4,971 - Mouser: 24 - Arrow: 150 - Future Electron.: 6500 - element14: 3,365
Reason: Specification matches with good stock quantity available from multiple distributors

Driver Description:

- Single chip LCD controller and driver
- Selectable backplane drive configuration: static or 2, 3, 4 backplane multiplexing
- 40 × 4-bit RAM for display data storage (25 x 3 used).
- Selectable display bias configuration: static, 1⁄2, or 1⁄3.
- Auto-incremented display data loading across device subaddress boundaries.
- Integrated Oscillator.
- Independent supplies possible for LCD and logic voltages.
- Wide power supply range: from 1.8 V to 5.5 V
- Wide logic LCD supply range:
    - From 2.5 V for low-threshold LCDs
    - Up to 6.5 V for high-threshold twisted nematic LCDs

- 400 kHz I2C-bus interface

Choice# 2 LCD Driver: BU9797AFUV-E2
IC Package: TSSOP-C48V
Protocol: 2-wire SDA, SCL
Link: BU9797AFUV-E2
Quantity Availability: DigiKey: 685 - Mouser: 15,996
Reason: Major specifications match with good stock quantity at Mouser.

Driver Description:
- 36 × 4-bit Integrated RAM for Display Data (25 x 3 used).
- 1⁄3 bias configuration.
- Integrated Oscillator.
- Independent supplies possible for LCD and logic voltages.
- Wide power supply range: from 2.5 V to 5.5 V
- Wide logic LCD supply range: from 0V to (VDD-2.4V)
- 2-wire Serial SDA SCL bus interface

Choice# 3 LCD Driver: PCF85176H/1,518
IC Package: TQFP-64
Protocol: I2C
Link: PCF85176H/1,518
Quantity: DigiKey: 7,168 - Mouser: 0 - Arrow: 1500
Reason: Major specifications match.

Driver Description:
- 40 × 4 universal LCD driver for low multiplex rates (25 x 3 will be used).
- Selectable display bias configuration: static, 1⁄2, or 1⁄3.
- Independent supplies possible for LCD and logic voltages.
- Wide power supply range: from 1.8 V to 5.5 V
- Wide logic LCD supply range:
    - From 2.5 V for low-threshold LCDs
    - Up to 6.5 V for high-threshold twisted nematic LCDs
- 400 kHz I2C-bus interface